

The Spatial Organization of Cold and Warm Temperature Synchrony

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Abstract

Spatial synchrony links local environmental variation to regional ecological stability, yet it is commonly reduced to a distance-decay curve or one local average. Such reductions can conceal the geographic arrangement of the relationships incident on a location. Using daily PRISM temperature from 1 November 2023 through 30 January 2024, we defined cold synchrony as the Spearman correlation of jointly selected lower-tail minimum temperatures and warm synchrony as the corresponding correlation of jointly selected upper-tail maximum temperatures. Their pairwise difference, $\Delta S = S_{\text{cold}} - S_{\text{warm}}$, is positive where cold synchrony is stronger and negative where warm synchrony is stronger. For each focal pixel, we retained the complete set of relationships as a local surface and evaluated its magnitude, heterogeneity, radial and directional organization, and dependence on spatial support. After bounded numerical validation, we applied the calculation across Colorado with a 100-km observation window. Warm synchrony was generally stronger and smoother than cold synchrony in this winter interval, and median ΔS was predominantly negative. Focal pixels with nearly identical median ΔS nevertheless had interquartile ranges of 0.242 and 0.062. Cumulative summaries at four radii reconstructed complete surfaces poorly, whereas radially adjusted directional profiles retained more structure. A separate 500-km pilot tested whether the field could instead be summarized by a focal-specific finite range. A common cold–warm range resolved at only one of 25 Colorado sites and none of five national representative sites. Continuous decay characteristics resolved more often, but no candidate met all prespecified stability criteria. Failure to identify one stable scalar range therefore did not indicate an absence of spatial structure. Cold and warm temperature synchrony formed distinct, geographically organized relationship fields whose interpretation depended on spatial support.

1 Introduction

Populations, communities, ecosystems, and environmental drivers often fluctuate together across space. This spatial synchrony matters because covariance among locations determines whether local variation cancels or accumulates at larger scales. Strong synchrony can increase regional variability and expose many locations to unfavorable conditions at the same time, including simultaneous declines that can elevate metapopulation extinction risk [Heino et al., 1997, Earn et al.,

2000]. Asynchrony can stabilize aggregate dynamics through portfolio effects, spatial insurance, or compensatory change, although the strength and mechanism of these effects depend on the system and scale considered [Earn et al., 2000, Loreau et al., 2003, Schindler et al., 2010]. Spatial synchrony is therefore central to questions about population persistence, community stability, regional risk, and ecological resilience [Bjørnstad et al., 1999a, Paradis et al., 1999, Liebhold et al., 2004].

Several processes can produce ecological synchrony. Dispersal connects local populations. Trophic interactions transmit fluctuations among species and locations. Shared environmental forcing can synchronize populations whose local dynamics respond similarly. The last mechanism is commonly associated with the Moran effect, which predicts that correlated environmental variation can generate correlated population fluctuations under appropriate dynamical assumptions [Moran, 1953, Liebhold et al., 2004]. The Moran effect does not imply that environment and populations have identical correlations, nor that correlated weather is the only source of synchrony. It does make the geography of environmental covariance an important part of the ecological problem.

Climate variables can remain correlated across large distances, and their synchrony commonly declines with separation [Koenig, 2002]. Weather synchrony can also change through time, altering the environmental conditions under which population synchrony is generated [Allstadt et al., 2015]. Increasing temperature synchrony has been observed alongside increasing synchrony in some North American bird populations [Koenig and Liebhold, 2016]. Long-term records also show that synchrony itself can change and can differ among temporal scales [Reuman et al., 2025]. These observations motivate a direct question: what does the environmental synchrony field itself look like? Answering that question requires more than estimating whether correlation decreases with distance. It requires asking whether the shape, spread, direction, and spatial support of climate relationships change across the landscape.

Synchrony can also differ between lower and upper portions of a distribution, and ecological work has shown that this asymmetry can itself have geographic structure [Walter et al., 2022]. Statistical extreme-value methods characterize joint-tail dependence by different constructions, including conditioning on one component being extreme [Heffernan and Tawn, 2004]; recent climate work has shown that warm- and cold-tail dependence can form distinct North American winter temperature patterns that are not equivalent to bulk correlation [Krock et al., 2023]. Our temperature contrast is related in motivation but differs in definition. It compares Spearman synchrony among jointly selected daily minimum temperatures at or below their local medians with synchrony among jointly selected daily maximum temperatures above their local medians. It is therefore a contrast between two temperature variables and two conditional day sets, not an extremal-dependence coefficient.

Several established approaches already retain more geography than a single isotropic distance-decay curve. In climatology, one-point correlation maps and full grid-cell correlation matrices have long been used to identify teleconnection patterns around selected reference locations [Wallace and Gutzler, 1981]. Climate-network analyses likewise represent grid cells as nodes and statistical dependence as edges in order to study the coupling architecture of climate fields [Tsonis and Roebber, 2004]. Because empirical climate networks can be sensitive to estimation and construction choices, their spatial features require explicit statistical validation [Haas et al., 2023]. In ecology, the geography-of-synchrony literature formalized the study of detailed pairwise patterns and introduced synchrony networks whose edges retain correlations among locations [Walter et al., 2017]. Subsequent work has combined distance decay, synchrony networks, matrix regression, and time-frequency analyses to resolve micro-geographic structure [Walter et al., 2021]. Other studies have produced regional synchrony maps by reducing each focal neighborhood to one statistic. For exam-

ple, Defriez and Reuman [2017] correlated each one-degree focal cell with cells within 500 km and averaged those correlations to map vegetation, temperature, and precipitation synchrony globally. Allen and Lockwood [2021] similarly mapped mean local population synchrony within 400 km and compared the result between historical periods. Together, these studies establish that synchrony has detailed geography, can be represented as a network, and can change through time and across spatial scales.

Geostatistical approaches address a related problem by allowing covariance structure to vary geographically. Nonstationary models for geophysical and climate fields can represent location-specific dependence ranges and geometric anisotropy rather than assuming one stationary isotropic covariance function [Jun and Stein, 2008, Kleiber and Nychka, 2012]. These models provide formal parametric descriptions of covariance. Our complementary aim is to retain and diagnose the conditional cold and warm relationship surfaces before choosing such a model or reducing each neighborhood to one parameter.

Those approaches provide the foundation for this study. Our contribution is not the first pairwise synchrony matrix, reference-location correlation map, network representation, or regional synchrony map. Instead, we make the unreduced local relationship surface the primary object of analysis. Regional means and network summaries are informative, but they do not necessarily describe the distribution and signed geographic arrangement of all relationships incident on a focal location. Retaining that arrangement makes it possible to test how much information is lost when the field is immediately reduced to one local value and whether synchrony strength, heterogeneity, direction, and spatial support describe distinct properties. This focus is especially relevant because synchrony can depend on temporal scale and geography, and its realized spatial scale can reflect more than distance alone [Bjørnstad et al., 1999b, Lande et al., 1999, Anderson et al., 2019].

Pairwise synchrony is the basic relationship. Holding one reference location fixed produces a map of its synchrony with the surrounding landscape. Repeating that map for many reference locations and then holding one focal location fixed reveals the complete set of relationships incident on that focal location. Indexing those values by the signed displacement, distance, and bearing of their comparison locations produces a local two-dimensional relationship surface. Mathematically, this surface is one row of a symmetric pairwise matrix; geographically, it retains spatial organization that an unordered vector or one local mean cannot provide.

This representation separates properties that are often conflated. Magnitude describes the typical synchrony of a focal location with its surroundings; heterogeneity describes how strongly that answer depends on the comparison location; and radial and directional structure describe how the relationships are arranged within the neighborhood. Spatial support is part of each quantity’s definition because admitting more distant relationships can change both magnitude and heterogeneity. These properties are related, but none is guaranteed to substitute for another.

We use daily PRISM temperature observations to ask four questions. First, do jointly cold and jointly warm conditions produce different synchrony fields? Second, what geographic information is concealed by reducing the relationships around a focal location to one median? Third, how do those relationships change with spatial support, and does distance alone describe their organization? Finally, can the field be reduced to a stable focal-specific finite range, or does its spatial organization require a richer description? We validate the pairwise and spatial calculations in bounded experiments before evaluating a 100-km neighborhood across Colorado and testing alternative spatial-scale summaries to 500 km. All empirical results refer to one winter window. They establish spatial organization within that interval, not a persistent climatic regime or temporal

trend.

2 Methods

2.1 Study system and PRISM climate data

We analyzed gridded daily temperature observations from the Parameter-elevation Regressions on Independent Slopes Model (PRISM). PRISM uses station observations and physiographic knowledge to map temperature and precipitation across the conterminous United States [Daly et al., 2008]. The synchrony calculation used daily minimum temperature (**tmin**) to represent lower-tail, relatively cold conditions and daily maximum temperature (**tmax**) to represent upper-tail, relatively warm conditions. We use “cold” and “warm” as concise labels for observations selected relative to each pixel’s distribution within the analyzed window. They are not absolute-event classifications or long-term climatological extremes.

The analysis used observations from 1 November 2023 through 30 January 2024. The implementation’s nominal `window_days=90` selects both endpoints and therefore contains 91 daily labels when the record is complete. We refer to this exact interval as the 91-day winter window. The primary geographic result retained focal pixels in Colorado and all eligible comparison pixels within a 100-km observation window. Comparison pixels outside the state were retained before focal outputs were clipped, so the political boundary did not truncate scientific neighborhoods. Bounded 20 by 20 and 100 by 100 pixel experiments were used for interpretation and numerical validation, respectively. Detailed data access, geometry, missingness, and provenance requirements are given in the Supplementary Information.

2.2 Conditional cold- and warm-tail synchrony

The method first identifies days on which both members of a pair occupy the same relative side of their own local temperature distributions, and then asks whether their ranked temperatures vary together on those days. It is therefore not a correlation of all winter observations, nor does it select days from one pixel and correlate them with an unconditioned series at the other pixel.

For a pair of pixels i and j , observations missing from either series were removed separately for **tmin** and **tmax**. Let $x_{i,t}^{\min}$ and $x_{j,t}^{\min}$ denote the paired minimum-temperature series. With split quantile $b = 0.5$, the lower-tail mask is

$$L_{ij,t} = \mathbb{I} \left[x_{i,t}^{\min} \leq Q_b(x_i^{\min}) \text{ and } x_{j,t}^{\min} \leq Q_b(x_j^{\min}) \right], \quad (1)$$

where each quantile is calculated from that series within the selected window after paired missing-value filtering. A pixel’s threshold can therefore differ among pairs when their missingness patterns differ. Cold-tail synchrony is the Spearman rank correlation among jointly selected observations,

$$S_{\text{cold}}(i, j) = \rho_S \left(x_{i,t}^{\min}, x_{j,t}^{\min} \mid L_{ij,t} = 1 \right). \quad (2)$$

For maximum temperature, the upper-tail mask uses a strict inequality,

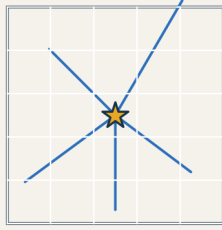
$$U_{ij,t} = \mathbb{I} \left[x_{i,t}^{\max} > Q_{1-b}(x_i^{\max}) \text{ and } x_{j,t}^{\max} > Q_{1-b}(x_j^{\max}) \right], \quad (3)$$

THE GEOGRAPHY OF CLIMATE SYNCHRONY

Relationships are retained as geographic fields before they are reduced to summaries

A ONE CENTER

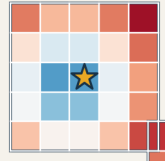
one reference center c



each line is $S(c, p)$

B MOVE THE CENTER

$M_c(p)$

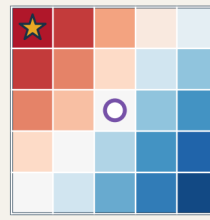


$M_{c+1}(p)$

each center produces a different landscape

C HOLD ONE FOCAL PIXEL

center origins contributing to $\text{Stack}(p)$



color = relationship with the focal pixel

RETAIN DISTINCT PROPERTIES



Conceptual illustration; colors are not observed values

Figure 1: The conceptual hierarchy. Pairwise synchrony $S(i, j)$ is the basic relationship. Holding center c fixed produces a center-reference landscape $M_c(p)$. Moving the center produces many landscapes. Holding focal pixel p fixed and collecting all eligible $S(c, p)$ values produces a geographically indexed relationship field. Mapping those values by signed displacement produces the local surface $L_p(\Delta x, \Delta y)$. Surface summaries describe magnitude, heterogeneity, radial structure, direction, and information not captured by compact summaries. Colors are conceptual and do not represent observations.

and

$$S_{\text{warm}}(i, j) = \rho_S \left(x_{i,t}^{\text{max}}, x_{j,t}^{\text{max}} \mid U_{ij,t} = 1 \right). \quad (4)$$

Spearman correlation was calculated from average ranks when ties occurred [Spearman, 1904]. A tail coefficient was returned only when at least the prespecified minimum number of jointly selected observations was available and both ranked series varied. Otherwise that tail coefficient was undefined. The bounded validation used a minimum of 10 selected observations; the final archived manifest must record the exact statewide setting and coverage rule.

We defined cold-warm synchrony asymmetry as

$$\Delta S(i, j) = S_{\text{cold}}(i, j) - S_{\text{warm}}(i, j). \quad (5)$$

Positive ΔS indicates stronger cold-tail synchrony. Negative ΔS indicates stronger warm-tail synchrony. The statistic compares two different variables and conditional subsets, so it is an asymmetry contrast rather than a single conventional correlation coefficient. It measures synchrony of relative local states: two pixels may occupy their respective cold or warm tails while having different absolute temperatures.

For a focal pixel, the mapped magnitude is the median of its valid pairwise contrasts,

$$\text{median}_c[\Delta S(c, p)].$$

It is not calculated as

$$\text{median}_c[S_{\text{cold}}(c, p)] - \text{median}_c[S_{\text{warm}}(c, p)],$$

which can give a different answer. All ΔS panels use one visual contract: red indicates negative ΔS and stronger warm-tail synchrony, near-white indicates zero, and blue indicates positive ΔS and stronger cold-tail synchrony. Cold- and warm-synchrony panels use their own correlation scales.

2.3 Center-reference landscapes and local synchrony surfaces

For a reference location c , pairwise synchrony with surrounding locations p defines a center-reference landscape,

$$M_c(p) = S(c, p). \quad (6)$$

We retained separate landscapes for $S_{\text{cold}}(c, p)$, $S_{\text{warm}}(c, p)$, and $\Delta S(c, p)$.

For every focal pixel p , we reversed the indexing and collected the values contributed by eligible centers. Each relationship retained its distance, bearing, signed displacement, and joint-tail sample count. Mapping the values by signed displacement produces

$$L_p(\Delta x, \Delta y; R) = S(c, p), \quad d(c, p) \leq R, \quad (7)$$

where R is the spatial support. A center-reference map therefore asks how one center relates to many focal pixels, whereas a local surface asks how one focal pixel relates to all centers around it. The pairwise statistic is symmetric, so each unique pair was calculated once and assigned to both endpoint surfaces with opposite signed displacement. The computational representation and self-pair rules for each product are documented in the Supplementary Information.

2.4 Primary spatial summaries

We described each focal surface with valid-pair count and coverage, the median as a measure of typical synchrony, and the IQR and median absolute deviation as robust measures of heterogeneity. We retained the full surface as the reference object. To describe its internal geography, we calculated 5-km annular profiles and angular profiles after removing the fine radial expectation. This adjustment prevented radial decay from being attributed to direction. We also compared the complete surfaces with cumulative summaries at 25, 50, 75, and 100 km and with several alternative compact representations. Those comparisons were diagnostic: no single composite synchrony index was defined.

2.5 Spatial support and finite-range tests

We first recalculated the focal median and IQR as comparison pixels were admitted in 5-km increments. These nested-radius curves describe sensitivity to the chosen observation window; they do not estimate the distance at which synchrony reaches a regional background. The statewide 100-km analysis should therefore be interpreted as a fixed local-scale control, not as an inferred universal range.

We then tested whether a focal-specific finite range could replace that fixed window. The pilot used the same winter interval, excluded self-pairs, extended discovery support to 500 km, and summarized newly encountered relationships in 20-km annuli. Annulus widths of 10 and 40 km and discovery limits from 200 to 500 km were used for sensitivity analysis. Cold and warm ranges were estimated independently. A candidate range required annular synchrony to approach an empirical distant background while the cumulative summary remained stable; a candidate near the discovery boundary was treated as censored rather than assigned the boundary value.

A common range was defined only when both tail-specific ranges resolved,

$$R_{\text{common}}^*(p) = \max\{R_{\text{cold}}^*(p), R_{\text{warm}}^*(p)\}. \quad (8)$$

Adaptive ΔS would then be the median pairwise contrast over the shared neighbor set within R_{common}^* ; unresolved ranges remained missing. The prespecified scale-up test used 25 spatially balanced Colorado sites and five national representative pixels. It required at least 70 percent Colorado coverage, at least 70 percent discovery stability, and resolution at four of five national sites.

Because a sharp endpoint rarely resolved, we also evaluated continuous properties of the same synchrony–distance curves: distances at which 25, 50, and 75 percent of estimated local excess synchrony had decayed, the positive area under the normalized excess curve (effective synchrony length), and an initial Theil–Sen slope over the first 100 km [Sen, 1968]. Candidate summaries had to meet coverage and representative-site criteria and remain stable to discovery support, annulus width, and background definition. Full definitions, status rules, and sensitivity grids are reported in the Supplementary Information.

2.6 Distance and directional structure

We evaluated distance and direction as descriptive organizers of each focal surface. The bounded diagnostic grouped distance and bearing explicitly; the Colorado representation used the 5-km

annular and radially adjusted angular profiles described above. A descriptive eta-squared statistic measured the fraction of surface variation associated with group means. It was used only to screen whether distance alone appeared sufficient, not as a causal variance partition. Distance, bearing, and unmeasured geographic factors can covary.

2.7 Validation and quality control

The pairwise and spatial calculations were checked at three levels. First, a 20 by 20 pixel block evaluated all 400 possible centers. Pair symmetry reduced 160,000 directed center-to-focal entries to 80,200 canonical pairs, and reconstructed cold, warm, and ΔS maps were compared with the validated center-reference implementation. Second, a 100 by 100 pixel benchmark compared tile-and-halo execution, endpoint reversal, chunking, and surface reconstruction before statewide application. Third, an end-to-end semantics audit followed observed temperatures through selection masks, rank correlations, pairwise contrasts, focal reductions, stored rasters, and display colors.

Five real relationships recomputed directly with SciPy agreed with stored values to a maximum absolute error of 2.22×10^{-16} , and three known-answer controls verified positive, negative, and zero ΔS . The audit found no sign reversal in the scientific calculation. Technical-report figures use the verified red-negative/warm and blue-positive/cold convention.

2.8 Colorado analysis and representation tests

After the 100 by 100 pixel benchmark passed, the fixed-radius calculation summarized 24,922,476 canonical relationships for Colorado focal pixels. The primary outputs were median cold synchrony, median warm synchrony, median pairwise ΔS , and robust spread of pairwise ΔS . Sixty-four complete local surfaces were retained to compare scalar summaries, cumulative radii, fine radial profiles, combined radial and directional profiles, low-order two-dimensional bases, and empirical principal-component representations. The sample was diagnostic rather than a probability sample for statewide inference. Computational architecture, runtime, software, fingerprints, and unresolved provenance fields are reported in the Supplementary Information and must be finalized in the immutable analysis archive.

3 Results

3.1 Cold and warm synchrony formed distinct relationship fields

The bounded experiment generated 400 center-reference landscapes from 80,200 canonical pairs. Reconstructed landscapes agreed exactly with the validated implementation for cold synchrony, warm synchrony, and ΔS . This established that the relational re-indexing preserved the underlying pairwise statistic.

In the worked example, a pixel 44.8 km from the center had cold synchrony 0.588 from 40 joint observations, warm synchrony 0.858 from 39 joint observations, and $\Delta S = -0.270$. Warm maximum temperatures were therefore more synchronous than cold minimum temperatures for that pair and winter window (figure 2).

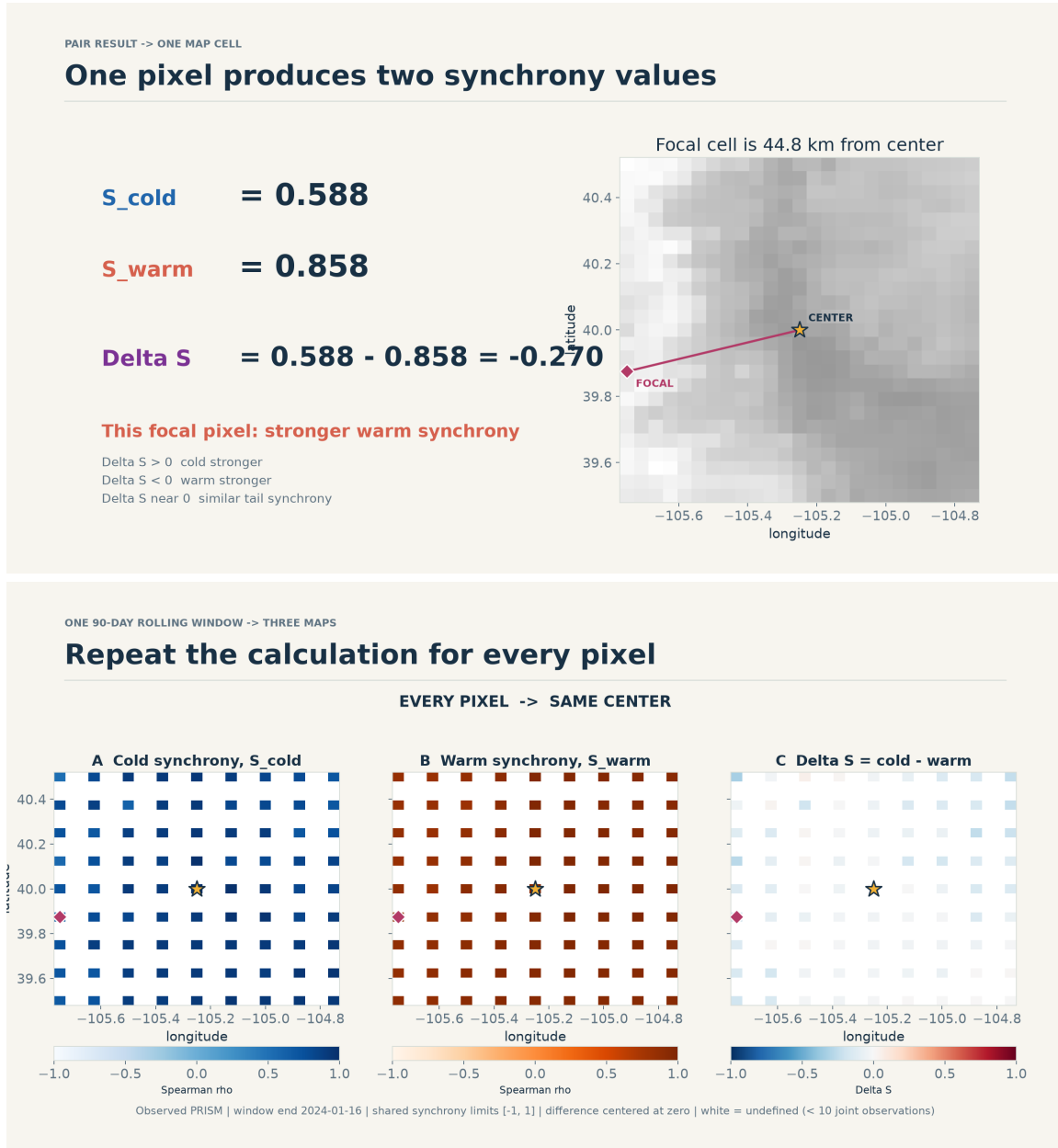


Figure 2: Validated center-reference synchrony from two rolling outputs. The upper panel reports the audited pair 44.8 km from the reference center over the 91 daily labels ending 30 January 2024. The lower panel is a separate illustrative map ending 16 January 2024 and shows joint lower-tail minimum- temperature synchrony, joint upper-tail maximum-temperature synchrony, and their pairwise difference for every focal pixel relative to that center. In the ΔS panel, red denotes stronger warm-tail synchrony and blue denotes stronger cold-tail synchrony. White in the supplied composite can represent either values near zero or masked pixels; numerical interpretation therefore uses the retained joint-count fields.

3.2 A focal median concealed heterogeneity and geographic organization

Local surface structure varied substantially across the experimental block. Focal (1, 5) had median ΔS approximately -0.063 and IQR 0.242, whereas focal (8, 14) had median approximately -0.062 and IQR 0.062. The underlying medians differed by only 0.0015. A reduction to the median would therefore describe these locations as nearly identical, while the full surfaces showed that focal (1, 5) had a much broader dependence on center choice.

The high-IQR surface was not broad because of a few isolated values. Its pairwise contrasts were geographically organized: centers in different parts of the block contributed systematically different values. The low-IQR example was comparatively narrow and homogeneous. A focal location can therefore have a given median relationship with its surroundings while being either insensitive or highly sensitive to which comparison locations are included (figure 3).

3.3 Focal summaries depended on spatial support

The radius required for focal summaries to stabilize varied among focal locations. The radius diagnostic used a low-IQR focal at (17, 5), with ΔS IQR approximately 0.056, and the high-IQR focal at (1, 5), with IQR 0.242. The experimental median and IQR criteria settled at 15 km for focal (17, 5) but not until approximately 80 km for focal (1, 5). Across the block, the median stable radius ranged from 15 to 80 km, with a median of 35 km. The IQR stable radius ranged from 15 to 100 km, with a median of 40 km.

These radii are diagnostic outcomes rather than intrinsic scales. They depend on the window, statistic, tolerance, and available domain. They show that fixed-radius summaries respond differently as support expands, but they do not estimate the distance at which pairwise synchrony reaches a distant background.

3.4 A universal finite synchrony range was not supported

The stricter 500-km pilot did not support an adaptive-radius map. At the primary 20-km bin width, nine of 25 Colorado sites resolved a cold range and nine resolved a warm range, but the resolved locations generally differed. Only one site resolved both tails and therefore a common range. Thus 24 of 25 common estimates, or 96 percent, were censored or unresolved. None of the 25 sites stabilized between 400 and 500 km, and none of five national representative pixels resolved a common range at 500-km discovery support.

Observed common-range coverage was 4 percent, compared with the prespecified 70-percent requirement. Discovery-stable coverage was 0 percent, and national representative coverage was zero of five rather than the required four of five. These results did not justify adaptive statewide or national mapping. Unresolved sites were left missing rather than assigned a 500-km range.

Fractional decay was more identifiable than a finite horizon. At 500-km discovery support and 20-km bins, d_{50} resolved for both tails at all 25 Colorado sites and all five national representative sites, and effective length was calculable at every site. Resolution alone was insufficient: warm d_{50} was stable to discovery support at only 52 percent of Colorado sites, effective length was discovery stable at 12 percent of cold sites and 16 percent of warm sites, and cold initial slope was stable to annulus width at only 40 percent. No candidate met all prespecified criteria for both tails, so all

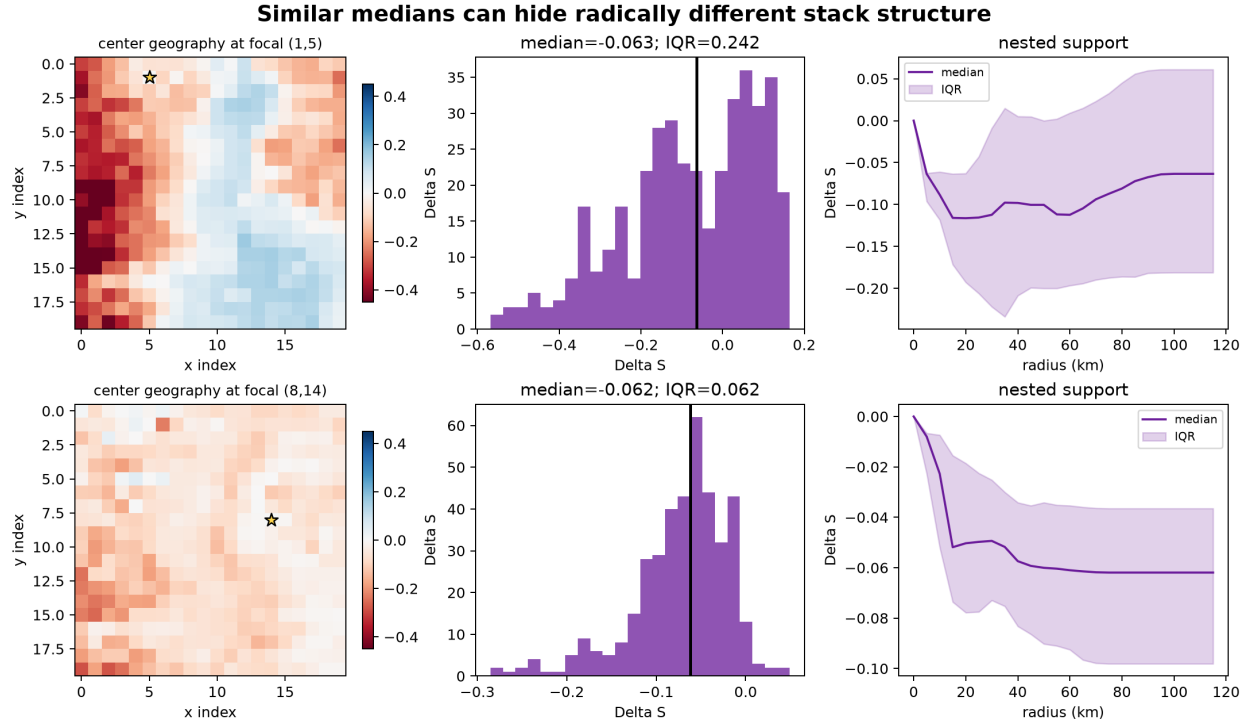


Figure 3: Similar median ΔS values concealed different local surfaces. Center-location maps, relationship histograms, and nested-support curves are shown for focal (1, 5) (top) and focal (8, 14) (bottom). Their median values were approximately -0.063 and -0.062 , but their IQRs were 0.242 and 0.062 . Thus similar central tendency arose from different relational geographies.

remained diagnostic rather than mapped products. The curves nevertheless showed that by 100 km the median fraction of local excess synchrony lost was 0.35 for cold and 0.25 for warm. Full coverage and stability results are reported in the Supplementary Information.

3.5 Distance alone did not describe local organization

The high-IQR focal surface exhibited both radial and directional organization. Descriptive grouped diagnostics gave eta-squared values of 0.104 for distance groups and 0.290 for direction groups, while the distance-residual IQR remained 0.185. Across all focal pixels, distance eta squared had a median near 0.10 and a maximum near 0.45; direction eta squared had a median near 0.285 and reached 0.629. These proportions are not an independent or causal variance partition, because distance, direction, and geographic context were not experimentally separated. They show that centers at similar distances were not exchangeable on distance alone and that bearing retained additional information.

Cold and warm components both varied across centers at the high-IQR focal. Cold synchrony had median 0.787 and IQR 0.218; warm synchrony had median 0.825 and IQR 0.175. Their pairwise contrast had median -0.063 and IQR 0.242. The broader cold component contributed more to ΔS heterogeneity, but the contrast could not be interpreted without retaining both inputs. Across the 64 complete Colorado surfaces, cumulative 25/50/75/100-km summaries reconstructed the full surfaces no better than a median-and-IQR baseline (report-defined normalized RMSE 1.02 and 1.01, respectively). A fine radial profile improved the error only slightly to 0.98, whereas adding a radially adjusted directional profile reduced it to 0.74. These diagnostic comparisons reinforce that spatial support and direction cannot be replaced by one focal scalar (figure 4).

3.6 Cold-warm asymmetry and heterogeneity were geographically structured

Cold- and warm-tail synchrony were positive across most of Colorado within the 100-km observation window, but their mapped spatial textures differed (figure 5). Warm synchrony was high and visually smoother over much of the state. Cold synchrony was more spatially variable, with lower-synchrony features in parts of central and south-central Colorado. Consequently, median pairwise ΔS was predominantly negative, indicating stronger warm-tail synchrony during this interval. The largest negative values formed spatially coherent features rather than isolated pixels. These are qualitative geographic descriptions of the reviewed maps. They do not attribute the patterns to topography or circulation, and they should not be interpreted as long-term temperature trends or fixed climate zones.

The spread of local relationships also varied geographically. Both ΔS IQR and median absolute deviation were higher in parts of central and southern Colorado and generally lower across much of the eastern plains. Their qualitative agreement indicates that the broad pattern was not specific to one robust dispersion statistic. Thus the geographic pattern was not limited to whether cold or warm synchrony was stronger; the variability of pairwise asymmetry within a focal neighborhood also formed a distinct surface.

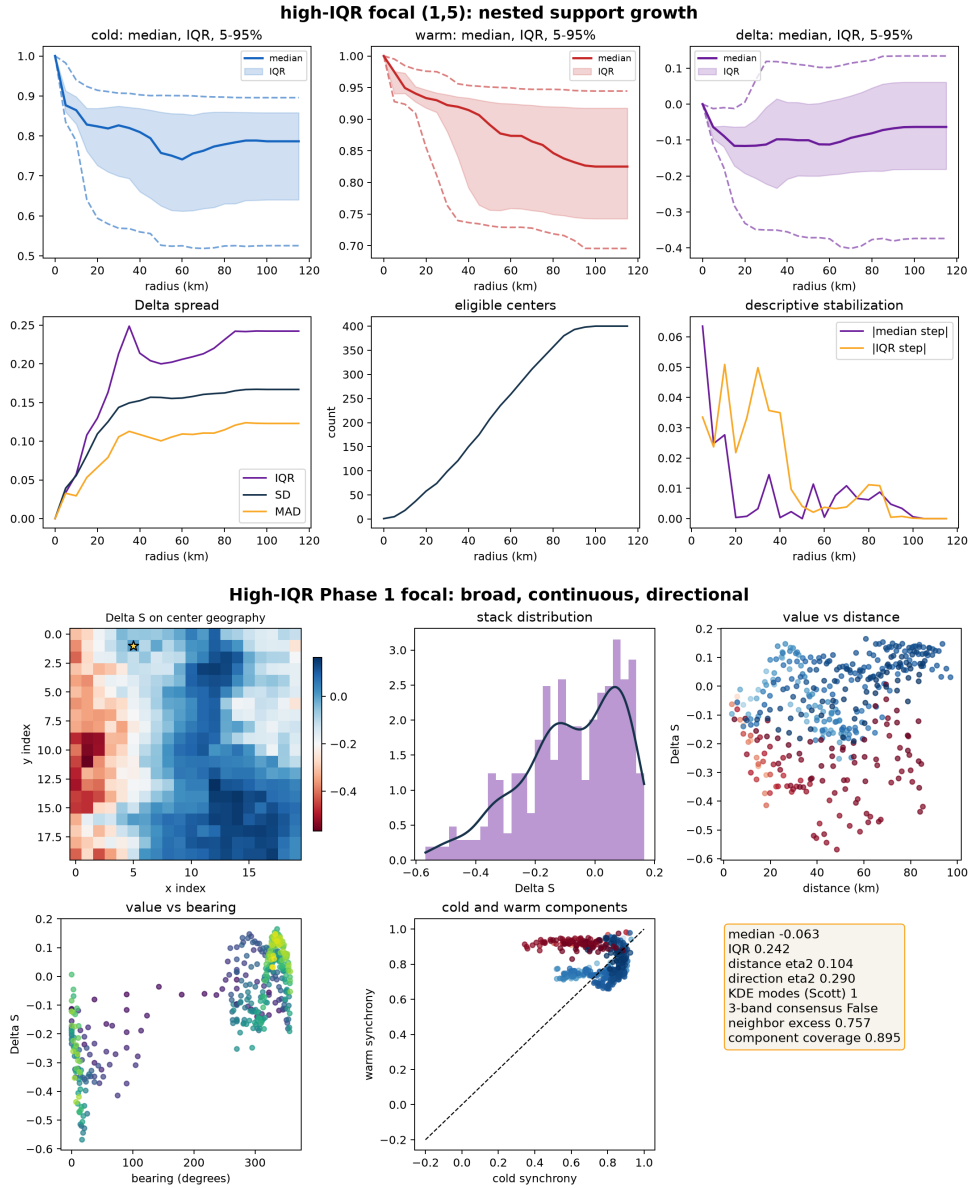


Figure 4: Spatial support and internal organization of the high-IQR focal surface at (1,5). The upper panel shows cold, warm, and ΔS summaries as increasingly distant centers are admitted. The lower panel maps ΔS by center origin and shows its distribution, distance relationship, bearing relationship, and cold-warm components. Distance and direction eta-squared values are descriptive and are not causal variance partitions.

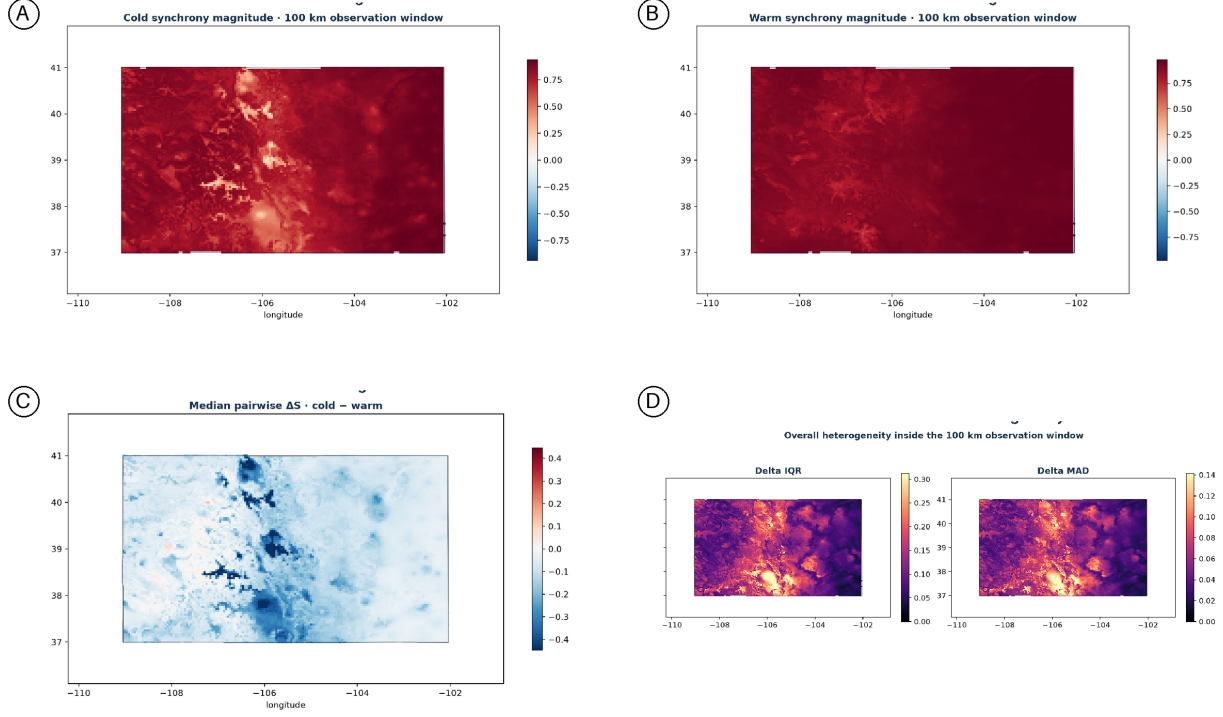


Figure 5: Colorado synchrony geography for the 1 November 2023 through 30 January 2024 window and a 100-km local observation window. Panels show median cold-tail synchrony (A), median warm-tail synchrony (B), median pairwise ΔS (C), and robust ΔS heterogeneity measured by IQR and median absolute deviation (D). Negative ΔS indicates stronger warm-tail synchrony. Panel C uses red for negative/warm stronger and blue for positive/cold stronger. The maps represent window-specific relationships, not long-term climate regions.

4 Discussion

4.1 Cold and warm synchrony are relational, not interchangeable

The main result is conceptual and empirical. A location does not possess one context-free synchrony value. It participates in a field of pairwise relationships, and that field changes as the reference location moves. The center-reference landscape and focal local surface are complementary views of the same symmetric pair calculations. One holds the center fixed and maps focal relationships. The other holds the focal location fixed and maps the origins of its relationships in signed local coordinates. This second view is mathematically a row of the pairwise matrix, but scientifically it is a spatial field with distance, direction, and geographic context. Retaining that field makes visible information removed when pairwise values are reduced immediately to a local mean, median, or isotropic distance-decay curve.

Cold and warm conditions also produced different fields in this interval. Warm synchrony was generally stronger and smoother across Colorado, while cold synchrony contributed more strongly to local heterogeneity in the focal example. Because the two coefficients use different temperature variables and jointly selected days, their contrast must be interpreted with both component fields and joint counts available. This is not a general claim that warm synchrony is always stronger; it is evidence that cold and warm synchrony were not interchangeable in the analyzed winter.

4.2 Magnitude and heterogeneity describe different properties

The paired focal examples provide the clearest evidence against immediate scalar reduction. Their median ΔS values differed by only about 0.0015, but their IQRs differed almost fourfold. A median described the typical cold-warm asymmetry but did not describe how strongly that answer depended on the reference center. Similar average synchrony can therefore arise from very different spatial relationships.

Surface heterogeneity should not be interpreted automatically as ecological response diversity, resilience, or instability. It is a property of the environmental relationship field. It may generate hypotheses about locations whose climatic connections vary strongly across nearby centers, but ecological consequences require biological observations and process models.

4.3 Spatial support is structured but not reducible to one resolved radius

The bounded diagnostics showed that local summaries changed at different rates as spatial support expanded. Their apparent stabilization points depended on the selected statistic and tolerance and should not be interpreted as the distance at which synchrony reaches a broad-scale background.

The representation comparison sharpened this result. Exact cumulative summaries at 25, 50, 75, and 100 km reconstructed the complete surfaces no better than a median-and-IQR baseline. The issue is not merely selecting the best one of four radii. Cumulative circles erase nonmonotonic radial structure and opposing values within the same ring. Fine annular profiles are more transparent, but the 100-km limit remains an observation window rather than a discovered scale.

The 500-km pilot made that distinction consequential. Tail-specific responses were often nonmono-

tonic or still reorganizing near the discovery boundary. Cold and warm ranges resolved at different locations, leaving only one of 25 Colorado sites with a common range and none of five national representatives. This negative result does not imply that synchrony has infinite range. It shows that one scalar range was not identifiable often enough under this window, criterion, and discovery support to justify adaptive maps. Replacing unresolved estimates with 500 km would confuse an observation boundary with a scientific estimate.

The follow-up analysis shows why endpoint failure should not be equated with an absence of decay. Half-decay distance resolved widely, effective length summarized the integrated curve, and initial slope described near-field change without requiring a background. Yet each metric exposed a different sensitivity: d_{50} to discovery support in the warm tail, L to the outer boundary in both tails, and cold β to annulus width. The fixed 100-km radius captured a meaningful part of local decline, but it was neither a universal half-decay distance nor a horizon. The contrast between relatively steep near-field decline and broad far-field tails suggests that a multiscale description may be more faithful than one adaptive cutoff.

Ecological synchrony studies have likewise shown that correlation length is a system-specific property and that the realized scale of population synchrony can combine environmental correlation with dispersal and density regulation [Bjørnstad et al., 1999b, Lande et al., 1999]. Our decay descriptors do not estimate those population mechanisms. They provide environmental quantities that could be compared with them in a coupled ecological analysis.

Synchrony also depends on temporal scale [Anderson et al., 2019, Reuman et al., 2025]. The present experiment isolates a spatial-support question within one winter window; it does not establish that the same distance structure applies in other seasons, years, variables, or temporal resolutions. The immediate conclusion is narrower: spatial support is an empirical dimension of the relationship field, but it did not collapse to one stable radius or map-ready curve statistic here.

4.4 Direction and geographic context retain information

Distance remains an important organizing variable. Nearby climate series often share more variation than distant series, and broad environmental synchrony can generate Moran-type population synchrony [Koenig, 2002]. The high-IQR focal example nevertheless showed that direction groups described more surface variation than distance groups under the exploratory diagnostics. Across the 64 Colorado surfaces retained for representation testing, adding a radially adjusted angular profile reduced the report-defined normalized reconstruction RMSE from 0.98 for a fine radial profile to 0.74. Thus distance alone did not preserve the sampled surface structure, and the directional signal was not confined to the original 20 by 20 block.

This does not show that direction causes synchrony. Bearing is a compact label for geography. It can encode topographic position, prevailing circulation, elevation transitions, land-water contrasts, observation geometry, or other spatially structured factors. The next analysis should combine directional summaries with explicit physiographic covariates rather than treating compass sector as a mechanism. The result does support a shift from strictly isotropic summaries toward spatially explicit models, consistent with geostatistical work that allows climate covariance ranges and anisotropy to vary by location [Jun and Stein, 2008, Kleiber and Nychka, 2012].

4.5 Ecological implications and boundaries

The classical Moran effect concerns population synchrony generated by correlated environmental forcing under assumptions about local dynamics [Moran, 1953]. Our analysis examines only the environmental side. It asks how correlated climate variation is organized and whether that organization is spatially homogeneous. We did not analyze population abundance, demographic rates, dispersal, trophic interactions, or biological response functions. Accordingly, this study does not test the Moran effect.

The results provide a richer environmental template for future Moran-type analyses. Spatial variation in environmental autocorrelation can alter the translation from forcing to population synchrony [Massie et al., 2015], and local response dynamics can amplify, dampen, or transform that environmental structure [Liebhold et al., 2004, Vasseur and Fox, 2009]. Populations separated by the same distance may therefore experience different covariance in forcing when their geographic contexts differ.

This template is also relevant to spatial insurance and portfolio effects, which depend on components not fluctuating in perfect synchrony [Loreau et al., 2003, Schindler et al., 2010]. These are connections, not quantities measured here. ΔS IQR does not measure biological response diversity, and a broad local surface does not establish resilience. Linking the environmental field to population or community time series will be necessary to test how climatic and biological sources of synchrony interact [Loreau and de Mazancourt, 2008].

4.6 Limitations and next tests

The current evidence comes from a single winter window. The Colorado analysis expands spatial coverage but cannot establish long-term climatic structure, seasonal generality, interannual stability, or national patterns. PRISM is an interpolated observational product with source and terrain-dependent uncertainty [Daly et al., 2008]. Tail selection uses relative medians, so results describe joint conditions within a window rather than absolute cold or heat extremes.

Descriptive distance and direction statistics do not separate mechanisms. Spatial autocorrelation and shared pair endpoints reduce the effective independence of pixel- and pair-level summaries [Dutilleul et al., 1993]. The 64 complete surfaces and 25 range-pilot sites were diagnostic samples rather than probability samples for statewide inference. The range and decay conclusions also depend on prespecified convergence, discovery-support, and bin-width criteria whose complete numerical configuration must accompany the archive.

An exploratory comparison with GHCN-Daily stations showed stronger agreement for cold synchrony and ΔS than for warm synchrony, but it was not an independent validation: stations were shared among edges, PRISM is informed by station observations, and the contributing-station roster was unavailable. Before claims are generalized, the calculation should be repeated across seasons and years, evaluated against alternative quantile and minimum-count settings, compared with unconditional and same-variable controls, and assessed with spatially and temporally appropriate uncertainty methods.

4.7 Temporal replication and historical reference

The present relative-tail definition is appropriate for asking how relatively cold and relatively warm conditions are organized within a period. Repeating that definition through time would reveal changes in within-period spatial organization, but every independently ranked period necessarily contains a lower and upper tail. It therefore cannot by itself determine whether temperatures that were unusual in an earlier climate are becoming more frequent.

A complementary future analysis could define a fixed, calendar-aware historical distribution $F_0(T \mid p, s)$ for location p and season or day of year s , and express later observations as $q_{\text{hist}}(p, t) = F_0[T(p, t) \mid p, s(t)]$. The historical reference would need to compare like seasons; a baseline period should be selected and tested in that future study rather than assumed here [Zhang et al., 2005, 2011]. This extension could distinguish changes in the frequency and magnitude of historically unusual temperatures from changes in their synchrony strength and spatial extent. Within-period and historically anchored tails would answer complementary questions, and comparing them could separate movement of the marginal temperature distribution from change in spatial dependence without, by itself, identifying cause. Spatial support and temporal reference are thus both parts of an extreme-temperature synchrony analysis. The present study addresses the spatial-support problem; historically anchored temporal analysis is the next extension.

5 Conclusion

Cold and warm temperature synchrony formed distinct geographic relationship fields in the analyzed winter. A focal median captured their typical asymmetry but not the heterogeneity, direction, or support dependence of the surrounding relationships. The field contained clear distance-dependent structure, yet a single finite cold–warm range was not robustly identifiable, and alternative decay scalars were not stable enough to map. Failure of one scalar summary was therefore not evidence of absent spatial organization. Treating synchrony strength and spatial structure explicitly provides a defensible basis for future temporal replication and for testing how relative and historically anchored temperature conditions differ through time.

Data and code availability

PRISM observations are provided by the PRISM Climate Group, Oregon State University. The climate-synchrony analysis code is currently developed within the CubeDynamics repository at <https://github.com/CU-ESIIIL/cubedynamics>. The final technical report will cite an immutable analysis archive and exact commit rather than presenting the general software repository as the scientific product.

ARCHIVE PLACEHOLDER: Insert DOI and immutable archive links for (1) the analysis code, (2) configuration and manifests, (3) bounded validation and diagnostic evidence, (4) statewide derived outputs permitted for redistribution, (5) empirical-range and decay-pilot configurations, statuses, and pair summaries, and (6) figure source data. Do not assign a repository DOI until an archive has actually been created.

Generative artificial intelligence transparency statement

During preparation of the technical report and Supplementary Information, members of the ASymClim Working Group used OpenAI Codex through the Prism research-writing environment on 24–25 September 2026. The environment identified the system as an OpenAI Codex agent based on GPT-5. It did not expose a more specific model snapshot or API model string, model weights or build identifier, system temperature, top-p value, random seed, or serving revision. Those unavailable values are therefore not reported, and the interaction is not expected to be reproducible word for word.

Codex was used interactively, under detailed author prompts, to read the technical report, Supplementary Information, bibliography, analysis reports, figure plans, and audit notes; propose and apply structural and line-level LaTeX edits; move supporting material from the main narrative to the supplement; improve prose, transitions, captions, and terminology; search for relevant literature; prepare and check BibTeX records; and run citation, consistency, and LaTeX-build checks. It also assisted with repository-native Python code for the two non-empirical conceptual diagrams. Codex did not generate PRISM or GHEN-Daily observations, replace the CubeDynamics calculations, perform the primary scientific analyses, or generate the empirical results reported in this paper. No model-generated numerical result was accepted as scientific evidence.

Scientific claims and numerical values were checked against the supplied analysis reports and retained artifacts. Candidate references were checked against primary or publisher records, and the final sources were audited for missing, unused, or duplicate citation keys and DOI values. The technical report and supplement were compiled after editing. The working group directed the scope of the work, decided which suggestions to retain, and remains responsible for the scientific interpretation, citation accuracy, wording, and final content. A more detailed record of inputs, tasks, exclusions, and verification appears in the Supplementary Information.

Author contributions

AUTHOR CONTRIBUTION PLACEHOLDER: Add the working-group membership, individual CRediT roles, and corresponding author after the group confirms the final authorship statement. ASymClim Working Group is currently listed as the group author.

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Competing interests

The authors declare no competing interests. This statement must be confirmed by all authors before submission.

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